



Please, Read Carefully!



Class Guidelines (First Session). Please,

-  **Mute your mic** when not speaking
-  **Keep your video off** unless requested
-  **Ask questions via chat** – I'll be watching!
-  **If I ask a question, please unmute and respond as soon as possible**
-  **Keep a paper & pen** ready for important notes
-  **Always feel free to ask relevant questions**
-  **You'll have time to speak at the end of the class**
-  **Recording & materials will be shared in Google Classroom within 30 minutes**

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Senior Business Intelligence Analyst
at Apple Gadgets Ltd.
Bachelor in Statistics, Islamic University



Let's talk about Data!



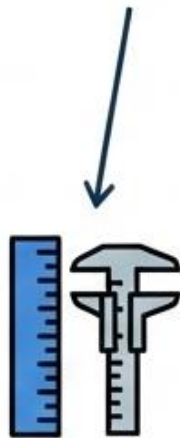
Data is a collection of **facts**, such as numbers, words, measurements, observations, images, videos, audio, or even just descriptions of things.



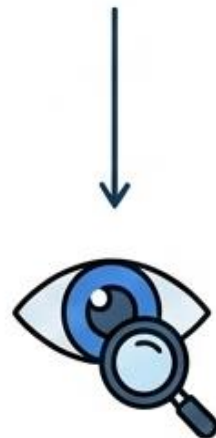
Numbers



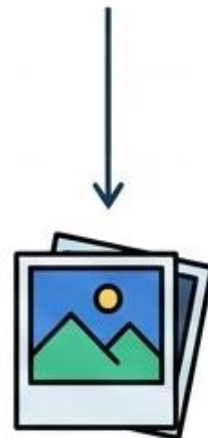
Words



Measurements



Observations



Images



Videos

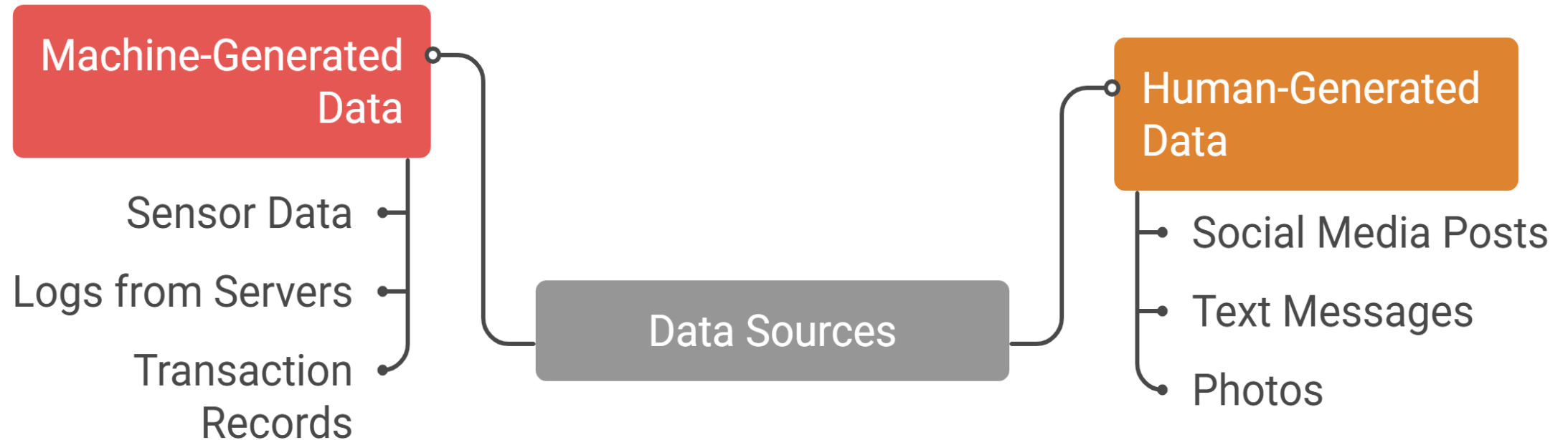


Audio



Descriptions
of things

Data Sources: Human vs. Machine



Data Structure Comparison

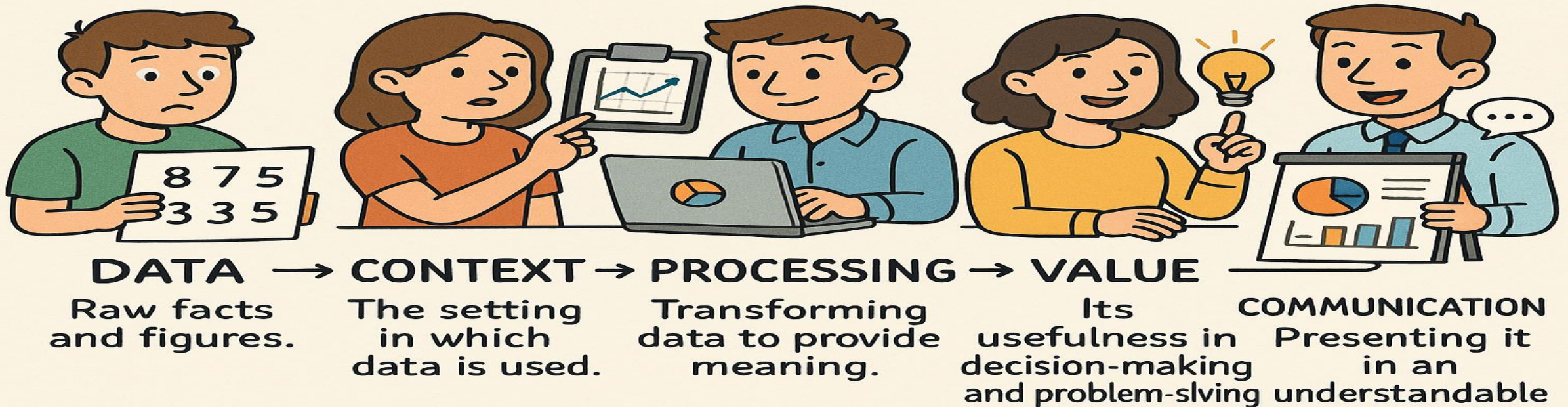
Characteristic	Structure	Storage	Examples
Structured Data	Tabular, fixed schema	Excel/CSV, SQL databases	Sales reports, production logs
Semi-structured Data	Flexible structure, keys, tags	NoSQL databases, data lakes	JSON, XML, log files
Unstructured Data	No fixed format or schema	File systems, object storage	Emails, images, videos

Data Types

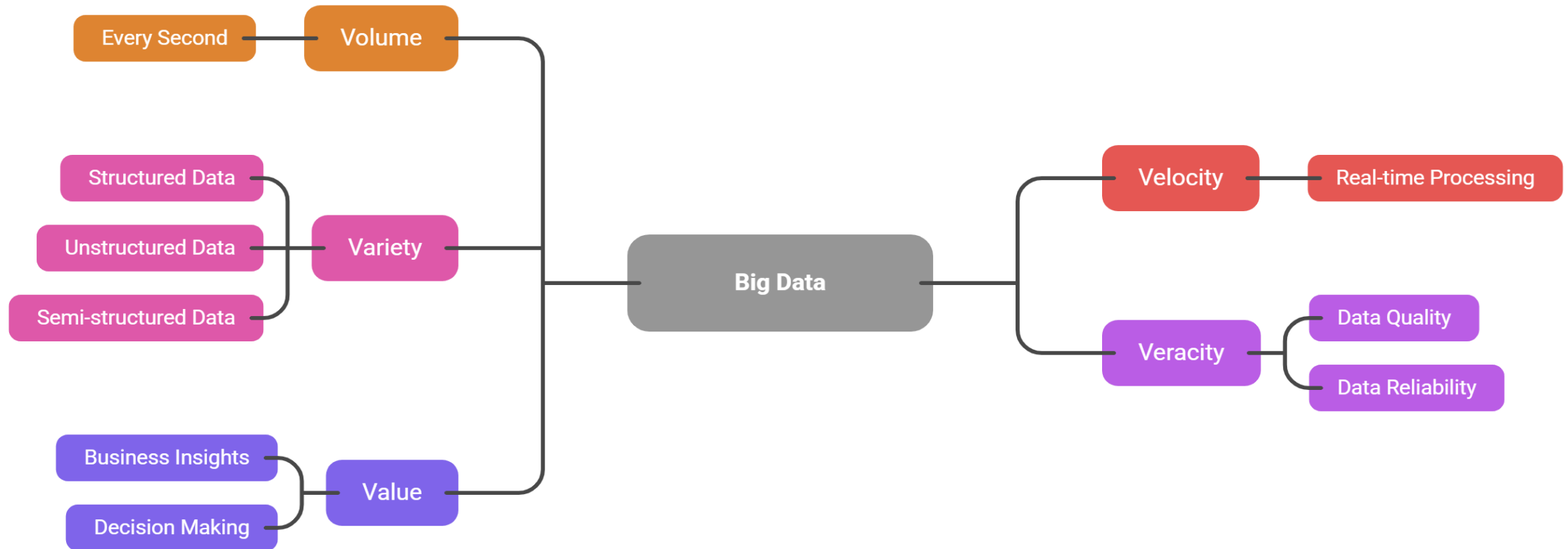
Characteristic	Data Type	Order	Examples
Nominal	Non-numeric, descriptive	No natural order	Gender, country, fabric type
Ordinal	Non-numeric, descriptive	Meaningful order	Size, quality rating, education level
Discrete	Numeric, countable	N/A	Number of orders, workers, defective pieces
Continuous	Numeric, measurable	N/A	Production time, fabric weight, cost

Information refers to data that has been processed, structured, or presented in each context to make it **meaningful and useful**. It is the result of interpreting, organizing, and structuring raw data in a way that provides value to the recipient.

THE PROCESS OF TRANSFORMING DATA INTO INFORMATION



Characteristics of Big Data (5V)



A Database is an organized collection of data, typically stored and accessed electronically from a computer system. Databases are designed to support the storage, retrieval, and management of data in a structured format.

Essential Database Features



Structured Data Storage

Data organized in tables, rows, and columns



Data Integrity

Ensures accuracy and consistency of data



Scalability

Handles increasing data and user loads



Security

Protects data with authentication and authorization

Types of Databases

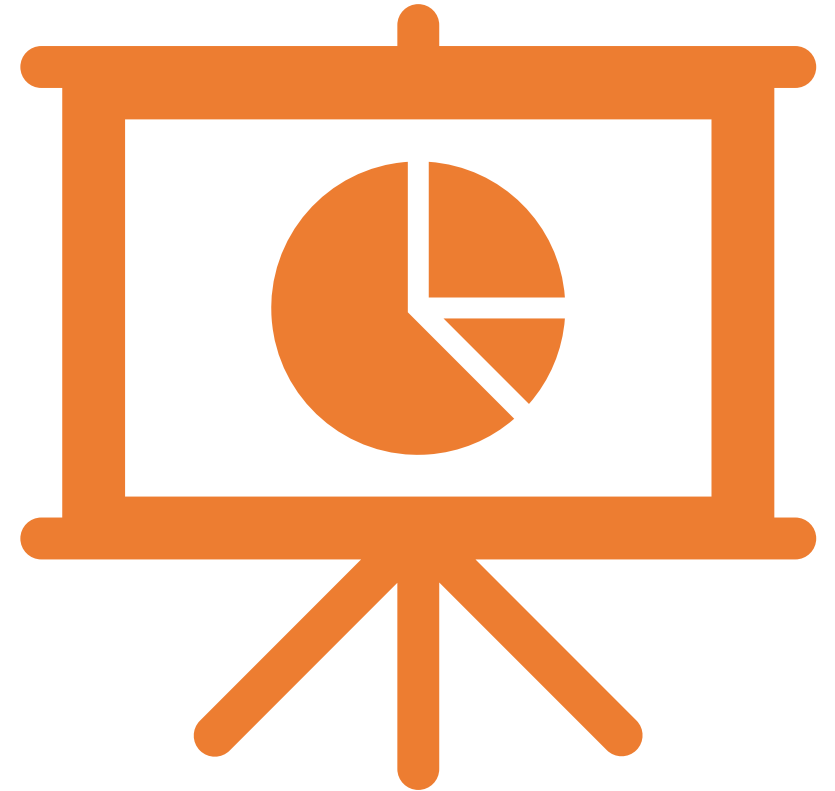
Characteristic	Relational (RDBMS)	NoSQL	In-Memory	Distributed
 Data Structure	Structured	Unstructured	In-Memory	Distributed
 Query Language	SQL	Varies	Varies	Varies
 Use Case	Transactional systems	Big data, real-time	Caching, session management	High availability, scalability
 Examples	MySQL, PostgreSQL, Oracle	MongoDB, Cassandra	Redis, Memcached	Apache Cassandra, Amazon DynamoDB

Data Analysis Steps!

These **seven steps** are commonly followed in data analysis projects to ensure a systematic approach from identifying the problem to delivering insights and recommendations.

1. Understanding the Business Problem
2. Data Collection
3. Data Cleaning and Preparation
4. Statistical Analysis / Data Analysis
5. Visualize the Results using Tools / Related Charts
6. Interpret and communicate the results
7. Iterate and Improve Results

- ❖ **Analysis**, Primarily focused on understanding **past and present** data. It aims to identify patterns, trends, and anomalies by inspecting and transforming data. The goal is to provide descriptive and diagnostic insights.



What is Data Analytics?

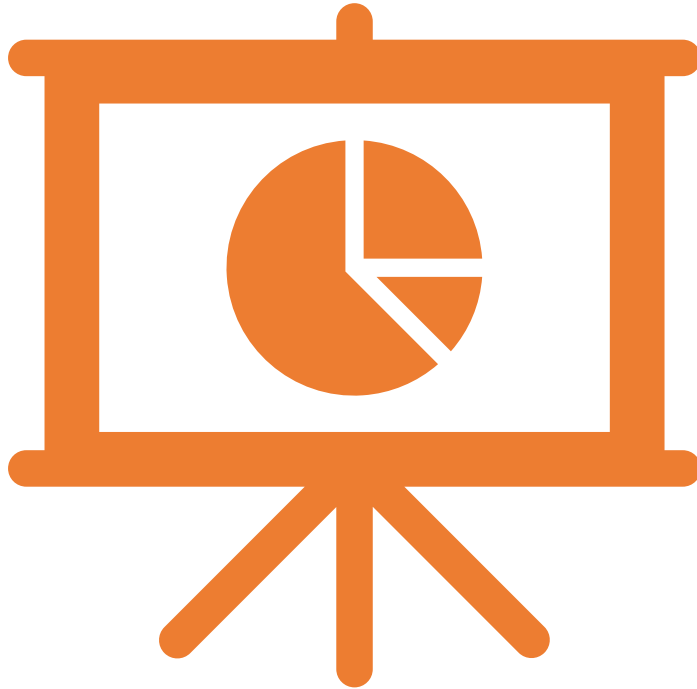
Definition

- ❖ **Analytics**, Broader in scope, encompassing the use of data to not only understand the past and present but also to **predict future** trends and outcomes using machine learning. It aims to provide predictive and prescriptive insights to support data-driven decision-making.



Data Analysis Vs. Data Analytics

Differences



Data Analysis

VS



Data Analytics

Descriptive

What happened? Summarizing historical data.

Example: Monthly sales reports, website traffic analysis.

Diagnostic

Why did it happen? Identifying causes of past events.

Example: Analyzing reasons for a drop in sales.

Predictive

What could happen? Forecasting future trends.

Example: Predicting customer churn or stock prices.

Prescriptive

What should we do? Recommending actions based on data.

Example: Optimizing marketing campaigns or supply chain management.

What Does a Data Analyst Do?

What is a Data Analyst?

Required skills and responsibilities

Key Aspects	Details
Main Role	☐ Interpreting data to provide actionable insights and business decisions.
Responsibilities	☐ Collecting and acquiring data ☐ Cleaning and preprocessing data ☐ Analysing data and identifying trends ☐ Visualizing data ☐ Reporting findings ☐ Collaborating with teams ☐ Developing predictive models ☐ Generating business insights ☐ Ensuring data governance and compliance
Required Skills	☐ Statistical analysis ☐ Data visualization ☐ Programming (Python, R, SQL) ☐ Data wrangling and cleaning ☐ Communication and reporting ☐ Machine learning basics

What is a Data Analyst?

Required skills and responsibilities

Key Aspects	Details
Tools and Technologies	☐ Data Analysis: Python (NumPy, Pandas, Matplotlib, Seaborn), R(dplyr), Excel ☐ Data Visualization: Tableau, Power BI, Matplotlib, Seaborn, ggplot2 ☐ Databases: SQL, MySQL, PostgreSQL ☐ Data Cleaning: Python (Pandas), Excel, ☐ Reporting: Tableau, Power BI, Looker Studio ☐ Machine Learning: Scikit-learn (Recommended)
Focus	☐ Providing actionable business insights
Output	☐ Comprehensive data reports ☐ Visual dashboards ☐ Predictive analytics models

Data analysts typically focus on **collecting, cleaning, and analyzing** data to answer specific questions or solve specific problems. They may use a variety of data analysis techniques, such as statistical analysis, data mining, and machine learning. They typically work with smaller datasets and focus on more detailed analysis.

Key skills and responsibilities:

- ❖ Strong analytical skills
- ❖ Proficiency in data analysis tools and techniques
- ❖ Domain knowledge
- ❖ Basic knowledge of business
- ❖ Excellent communication skills

Data analysts are entry-level professionals. The median salary for **data analysts** is \$70,000 per year. data analysts with more experience and expertise can earn significantly more.

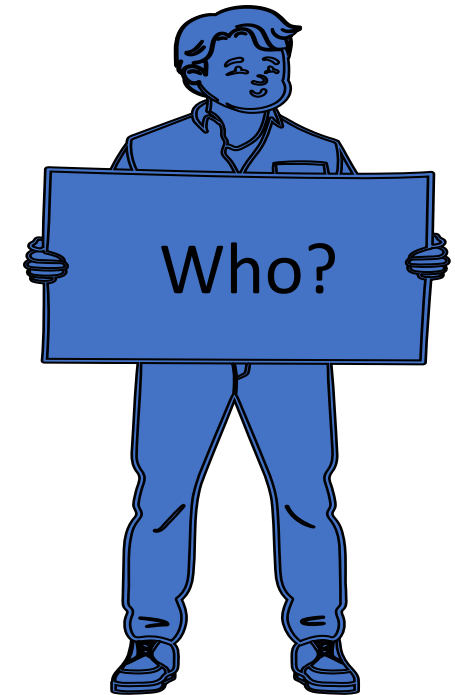


BI analysts typically focus on **creating** reports, dashboards, and visualizations that communicate data insights to stakeholders. They may use a variety of BI tools, such as Power BI, Tableau, and Qlik Sense. They typically work with larger datasets and focus on providing insights that can be used to make business decisions.

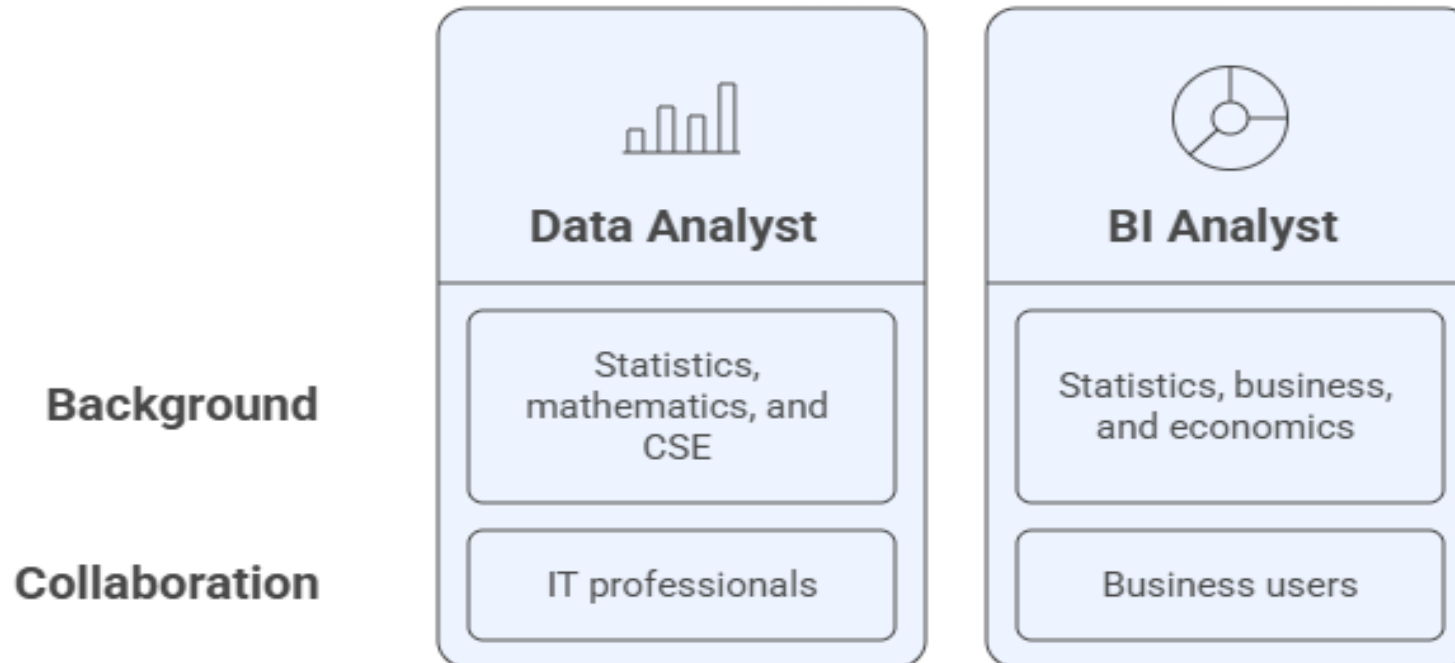
Key skills and responsibilities:

- ❖ Strong analytical skills
- ❖ Proficiency in data analysis tools and techniques
- ❖ Domain knowledge
- ❖ Strong knowledge of business
- ❖ Excellent communication skills

BI analysts are well-paid professionals. The **median salary** for BI analysts is \$90,000 per year. BI analysts with more experience and expertise can earn significantly more.



Data Analyst vs. BI Analyst

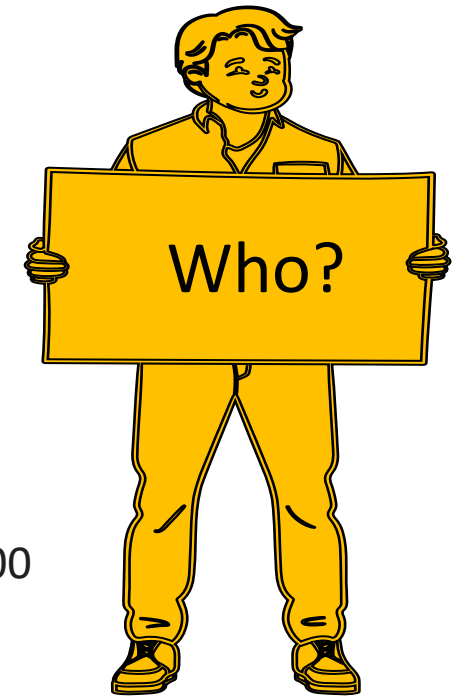


Data scientists are experts in using machine learning and artificial intelligence to analyze data. They develop predictive models that can be used to make informed decisions.

Key skills and responsibilities:

- ❖ Strong analytical skills
- ❖ Proficiency in programming languages like SQL and Python.
- ❖ Proficiency in data analysis tools and techniques
- ❖ Enough statistical and mathematical knowledge
- ❖ Strong knowledge of **machine learning, deep learning, and AI**
- ❖ Excellent communication skills

Data scientists are high-paid professionals. The **median salary** for data scientists is \$130,000 per year. Data scientists with more experience and expertise can earn significantly more.



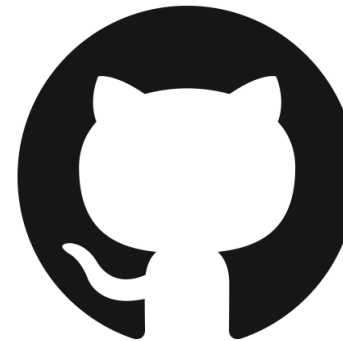
Importance of Social Networking & Sharing



www.linkedin.com

kaggle

www.kaggle.com



www.github.com



Educational
Facebook Groups

What are we going to learn??

Let's visit the course structure!

ASK Me any Questions!